

Solved Past Paper BS-AD Semester 8 (Electroanalysis Method-II) Chem-464

Re-Conduct Paper:

1. What is voltammetric cell?

A voltammetric cell is an electrochemical cell used to measure the current-voltage relationship of a solution or a substance. It's commonly used in analytical chemistry to study the electrochemical behavior of various species.

A typical voltammetric cell consists of three electrodes:

- **Working electrode (WE):** where the electrochemical reaction occurs
- **Reference electrode (RE):** provides a constant potential reference point
- **Counter electrode (CE):** also known as the auxiliary electrode, it completes the circuit and allows current to flow

2. What is stirring effect on the amperometry performance?

Stirring can significantly impact amperometry performance in several ways:

- **Mass transport enhancement:** Stirring increases the transport of analyte molecules to the electrode surface, leading to a higher current response.
- **Convective diffusion:** Stirring introduces convective forces that enhance the diffusion of species towards the electrode, increasing the current response.
- **Reduced fouling:** Stirring helps remove reaction products or adsorbed species from the electrode surface, reducing fouling and maintaining electrode activity.
- **Improved reproducibility:** Stirring minimizes variations in current response caused by changes in solution dynamics.

3. What is residual current in polarography?

In **polarography**, **residual current** is the background current that flows even without the presence of any electroactive species. It has two components:

Capacitive (Charging) Current: Due to the charging of the electrical double layer at the electrode surface.

Faradaic Current: From minor redox reactions caused by impurities in the solution.

4. What are the different types of amperometry?

There are several types of **amperometry**, each designed for specific electrochemical applications.

- Constant Potential Amperometry (CPA)
- Pulsed Amperometry (PA)
- Differential Pulse Amperometry (DPA)
- Cyclic Amperometry
- Multi-Pulse Amperometry
- Hydrodynamic Amperometry

5. What is the advantage of solid electrode (like platinum) over mercury electrode in cyclic voltammetry?

The advantage of a solid electrode, like platinum, over a mercury electrode in cyclic voltammetry includes:

1. **Wider Potential Window:** Solid electrodes like platinum can operate over a broader potential range compared to mercury, allowing for a wider range of redox reactions to be studied.
2. **Stability and Durability:** Platinum is chemically and physically stable, whereas mercury can oxidize or evaporate, making solid electrodes more durable and easier to handle.
3. **Safety:** Platinum is non-toxic and environmentally friendly, while mercury poses health and environmental risks.

6. Why cyclic voltammetry has been recognized as the most versatile electrochemical technique?

Cyclic voltammetry is highly versatile because it allows the exploration of redox reactions, distinguishes between reversible and irreversible processes, and provides both kinetic and thermodynamic data. It can be used for qualitative and quantitative analysis, studies of electrode surfaces, and is applicable to various chemical systems, making it a powerful tool in electrochemistry.

7. What are three mechanisms of mass transport in voltammetry?

In voltammetry, mass transport refers to the movement of electroactive species towards or away from the electrode surface. The three main mechanisms of mass transport are:

- **Diffusion:** The random motion of molecules due to concentration gradients, resulting in the transport of electroactive species towards or away from the electrode surface.
- **Migration:** The movement of charged species (ions) in an electric field, which can enhance or hinder mass transport, depending on the direction of the electric field and the charge of the species.
- **Convection:** The transport of electroactive species through fluid motion, such as stirring, agitation, or flow, which can significantly enhance mass transport and increase the current response.

8. What is the role of inert supporting electrolyte in polarography?

In polarography, an inert supporting electrolyte minimizes the migration of electroactive species, improves solution conductivity, stabilizes the electrochemical environment, and prevents interference by being non-reactive. It also helps maintain a uniform electric field, leading to more accurate measurements and better resolution of electrochemical signals during the experiment.

Example:

- An inert supporting electrolyte used in polarography is **potassium chloride (KCl)**. KCl is commonly used because it is a strong electrolyte that provides a stable ionic environment without participating in the electrochemical reactions being studied.

9. What do you mean by polarogram?

A polarogram is a graphical representation of the current response as a function of the applied potential in polarography. It plots the measured current on the **y-axis** against the electrode potential on the **x-axis**. The resulting curve reveals information about the electrochemical behavior of the analyte, such as reduction or oxidation peaks, which can be used to identify and quantify the substance being analyzed.

10. Give any two advantages of amperometry.

Two advantages of amperometry are:

1. **High Sensitivity:** Amperometry can detect very low concentrations of analytes due to its ability to measure small changes in current, making it highly sensitive for various analytical applications.
2. **Rapid Analysis:** The technique allows for quick and continuous monitoring of electrochemical reactions, providing real-time data and enabling fast analysis compared to some other methods.

11. Give the difference between linear sweep voltammetry and cyclic voltammetry?

Linear Sweep Voltammetry	Cyclic Voltammetry
A single sweep of potential from an initial to a final value	Multiple sweeps of potential between two fixed values (lower and upper potential limits)
No reversal of potential direction	Reversal of potential direction after reaching the switching potential
Only one cycle is performed	Multiple cycles are performed

12. What are the advantages of polarography?

High Sensitivity: It can detect very low concentrations of analytes due to its ability to measure small currents, making it highly sensitive for trace analysis.

Quantitative Analysis: Polarography provides quantitative data by correlating current responses to analyte concentrations, allowing for accurate measurement.

13. Discuss the deficiencies of amperometry.

Amperometry can suffer from interference from other electroactive species, non-faradic background currents, limited selectivity, electrode fouling, signal drift, and reagent consumption. These issues can affect the accuracy and precision of measurements, requiring careful calibration and maintenance to ensure reliable results.

14. Why is Ag/AgCl used as reference electrode in amperometry?

The Ag/AgCl reference electrode is used in amperometry because:

1. **Stable Potential:** It provides a stable and well-defined reference potential, essential for accurate and reproducible measurements.
2. **Reproducibility:** The electrode's potential is consistent and does not change significantly over time, ensuring reliable results.
3. **Ease of Use:** It is straightforward to prepare and use, making it practical for various electrochemical setups.
4. **Non-reactivity:** The Ag/AgCl electrode does not interfere with the electrochemical reactions of interest, allowing for accurate measurement of the analyte's response.

15. How oxygen interferes in polarographic analysis?

Oxygen interferes in polarographic analysis by:

1. **Reduction Reaction:** Oxygen undergoes reduction at the electrode, producing a current that overlaps with the analyte's signal, leading to an unwanted background current.
2. **Competing with Analytes:** It competes with the analyte for reduction at the electrode, distorting the polarogram and affecting the accuracy of the analysis.

Long Questions:

Answer the following questions.

Q. No. 2:

- (a) Write down the basic principle of amperometry.
- (b) Discuss the applications of amperometry.

Q. No. 3:

- (a) Write note on qualitative aspects of polarographic analysis.
- (b) Briefly, Discuss the instrumentation of polarography.

Q. No. 4:

(6) Explain in detail anodic stripping voltammetry.

(b) Write down applications of voltammetry.

Annual Paper:

1. What is the basic mechanism of mass transport in voltammetry?

The basic mechanisms of mass transport in voltammetry are:

- **Diffusion:** The random motion of molecules due to concentration gradients, resulting in the transport of electroactive species towards or away from the electrode surface.
- **Migration:** The movement of charged species (ions) in an electric field, which can enhance or hinder mass transport, depending on the direction of the electric field and the charge of the species.
- **Convection:** The transport of electroactive species through fluid motion, such as stirring, agitation, or flow, which can significantly enhance mass transport and increase the current response.

2. Differentiate Between Diffusion current and Migration current.

Diffusion Current	Migration Current.
Caused by the random motion of ions or molecules in a solution	Caused by the movement of ions in a solution under the influence of an electric field
Driven by concentration gradients	Driven by the electric field strength
Ions or molecules move from an area of higher concentration to an area of lower concentration	Ions move in response to the electric field, not due to concentration gradients
No external force is applied	External force (electric field) is applied
Current is proportional to the concentration gradient	Current is proportional to the electric field strength and ionic mobility

3. What is the role of conventional and faradaic current in voltametric measurements?

Faradaic Current:

- **Role:** Directly related to the electrochemical reactions occurring at the electrode surface. It results from the redox processes of the analyte, where the current is proportional to the rate of these electrochemical reactions.

Provides information about the concentration of the analyte and the kinetics of the redox reactions. It is the primary signal of interest in voltammetry.

Conventional Current (or Non-Faradaic Current):

- **Role:** Includes all currents not directly related to the redox reactions of the analyte.

This can include capacitive currents, background currents, and leakage currents.

Acts as noise or interference that can obscure the faradaic current.

Understanding and minimizing conventional current is crucial for accurate measurement of the faradaic current.

4. Describe two disadvantages of Amperometry.

Two disadvantages of amperometry are:

1. **Interference:** Other electroactive substances in the sample can interfere with the measurement, leading to inaccurate results if not properly accounted for.
2. **Electrode Fouling:** The electrode surface can become fouled or contaminated over time, affecting the accuracy and reproducibility of the measurements.

5. Why Mercury electrode is widely used?

Mercury electrodes are widely used because they offer a broad potential range, are chemically inert, and provide a renewable, smooth surface for electrochemical measurements. They allow for sensitive detection of trace metals, form amalgams with various metals, and produce well-defined results, making them ideal for analytical applications.

6. How potentiometric titration is useful for determination of indicator for endpoint of titration?

Potentiometric titration helps determine the endpoint without a visual indicator by measuring the potential change in the solution as titrant is added. It provides accurate detection of the equivalence point, even in cases where visual indicators may not work, like in colored or turbid solutions. It ensures precision by identifying the sudden change in voltage that occurs at the endpoint.

7. What is voltammogram?

A voltammogram is a graphical representation of the current response as a function of the applied potential during a voltammetric experiment. It plots the current (y-axis) against the electrode potential (x-axis) and provides information about the redox behavior of electroactive species, including the identification of peaks corresponding to oxidation or reduction processes.

8. How voltametric techniques are classified?

Voltammetric techniques are classified by how potential and current are handled:

1. **LSV**: Linear potential sweep, current measured.
2. **CV**: Forward and reverse sweeps, studying redox.
3. **Pulse Voltammetry**: Potential pulses, current measured (e.g., DPV).
4. **SWV**: Square wave pulses, high sensitivity.
5. **Chronoamperometry**: Constant potential, current over time.

9. Define migration current.

Migration current is the movement of ions in a solution under the influence of an electric field.

Migration current occurs when:

- An electric field is applied to a solution containing ions.
- Ions move in response to the electric field, with cations (positively charged ions) moving towards the cathode (negative electrode) and anions (negatively charged ions) moving towards the anode (positive electrode).
- The movement of ions creates an electric current, known as the migration current.

Long Question:

1. Explain in detail the anodic stripping Voltammetry.
2. Write down application of Voltammetry.
3. Explain in detail the four types of potentiometric titration.
4. Describe application advantages and disadvantages of Amperemeter.

BS-Regular Paper

1. How Oxygen interferes in polarographic analysis?

Oxygen interferes in polarographic analysis by:

- **Reduction Reaction:** Oxygen undergoes reduction at the electrode, producing a current that overlaps with the analyte's signal, leading to an unwanted background current.
- **Competing with Analytes:** It competes with the analyte for reduction at the electrode, distorting the polarogram and affecting the accuracy of the analysis.

2. What do you know about residual current in polarography?

In **polarography**, **residual current** is the background current that flows even without the presence of any electroactive species. It has two components:

Capacitive (Charging) Current: Due to the charging of the electrical double layer at the electrode surface.

Faradaic Current: From minor redox reactions caused by impurities in the solution.

3. What are the limitations of Amperometry?

Amperometry has several limitations, including:

1. **Interference:** Other electroactive substances in the sample can interfere with the measurement, leading to inaccurate results if not properly accounted for.
2. **Electrode Fouling:** The electrode surface can become fouled or contaminated over time, affecting the accuracy and reproducibility of the measurements.

4. What is the basic principle of Anodic stripping voltammetry?

The basic principle of Anodic Stripping Voltammetry (ASV) involves three main steps:

- **Accumulation:** The analyte is deposited onto the electrode surface (usually a mercury or gold electrode) at a reduced potential, allowing the analyte to accumulate onto the electrode.

- **Stripping:** The potential is then reversed and the analyte is stripped (oxidized) from the electrode surface, resulting in a current peak.
- **Measurement:** The stripping peak current is measured and related to the concentration of the analyte in the sample.

5. What do you mean by Cyclic Voltammetry?

Cyclic Voltammetry (CV) is an electrochemical technique where the potential applied to a working electrode is cycled linearly forward and backward between two set values, while the resulting current is measured. This process allows the study of redox reactions, providing information about the electrochemical properties of a substance, such as reaction reversibility, redox potentials, and the kinetics of electron transfer. CV is commonly used for characterizing materials, understanding reaction mechanisms, and studying the behavior of electroactive species.

6. What is the basic mechanism of mass transport in voltammetry?

The basic mechanisms of mass transport in voltammetry are:

- **Diffusion:** The random motion of molecules due to concentration gradients, resulting in the transport of electroactive species towards or away from the electrode surface.
- **Migration:** The movement of charged species (ions) in an electric field, which can enhance or hinder mass transport, depending on the direction of the electric field and the charge of the species.
- **Convection:** The transport of electroactive species through fluid motion, such as stirring, agitation, or flow, which can significantly enhance mass transport and increase the current response.

7. Differentiate between anodic current and cathodic current in polarography.

Anodic Current	Cathodic Current
The current measured when the working electrode is at a positive potential relative to the reference electrode.	The current measured when the working electrode is at a negative potential relative to the reference electrode.

Involves oxidation reactions at the electrode surface, where the analyte loses electrons.	Involves reduction reactions at the electrode surface, where the analyte gains electrons.
Represents the current due to the oxidation of species in the solution.	Represents the current due to the reduction of species in the solution.

8. Describe the role of inert supporting electrolyte in polarography.

In polarography, an inert supporting electrolyte minimizes the migration of electroactive species, improves solution conductivity, stabilizes the electrochemical environment, and prevents interference by being non-reactive. It also helps maintain a uniform electric field, leading to more accurate measurements and better resolution of electrochemical signals during the experiment.

Example:

- An inert supporting electrolyte used in polarography is **potassium chloride (KCl)**. KCl is commonly used because it is a strong electrolyte that provides a stable ionic environment without participating in the electrochemical reactions being studied.

9. Write down the principle of Amperometry.

The principle of amperometry is based on measuring the current that flows through an electrochemical cell as a result of a constant applied potential. In amperometry, a fixed potential is applied between a working electrode and a reference electrode, and the resulting current, which is proportional to the concentration of the electroactive species in the solution, is measured. The current is directly related to the rate of the electrochemical reaction occurring at the working electrode, allowing for quantitative analysis of the analyte concentration.

10. Give disadvantages of Anodic stripping voltammetry.

Anodic stripping voltammetry (ASV) has some disadvantages, including:

1. **Complex Sample Preparation:** Sample preparation can be complicated, requiring steps like preconcentration and careful handling to avoid contamination or loss of analytes.

2. **Interference:** Presence of interfering substances in the sample can affect the accuracy of measurements by overlapping with the signals of target analytes.

11.Distinguish between migration current and residual current.

Residual current:	Migration Current.
Caused by the charging of the double layer capacitance at the electrode-solution interface	Caused by the movement of ions in a solution under the influence of an electric field
Resulting from the rearrangement of ions and dipoles in the double layer	Resulting from the application of an electric field
Not directly related to the analyte concentration	Directly related to the concentration of the analyte

12.Describe Amperometric titration involving electro-reducible vs electro-oxidizable species.

In amperometric titration:

1. **Electro-reducible Species:** The analyte gains electrons (is reduced) at the electrode. The endpoint is detected by a change in current as the reducing titrant is added.
2. **Electro-oxidizable Species:** The analyte loses electrons (is oxidized) at the electrode. The endpoint is identified by a change in current as the oxidizing titrant is added.

Long Questions:

1. What is meant by diffusion current? Discuss factors affecting it.
2. Discuss the application of Amperometry.
3. Discuss Anodic Stripping Voltammetry.
4. Discuss Various types of electrodes in polarography.
5. Discuss quantitative application of Anodic Stripping Voltammetry.
6. Discuss electrodeposition step and Voltammetric completion step in voltammetry.
7. What half wave potential. What information is obtained from polarographic curves.
8. Write a note on Qualitative aspect of polarographic analysis.

9. Discuss instrumentation of amperometric titration.
10. Discuss the construction and working of Voltammetric cell.
11. Polarography is applied to organic and inorganic substances. Explain in details.